

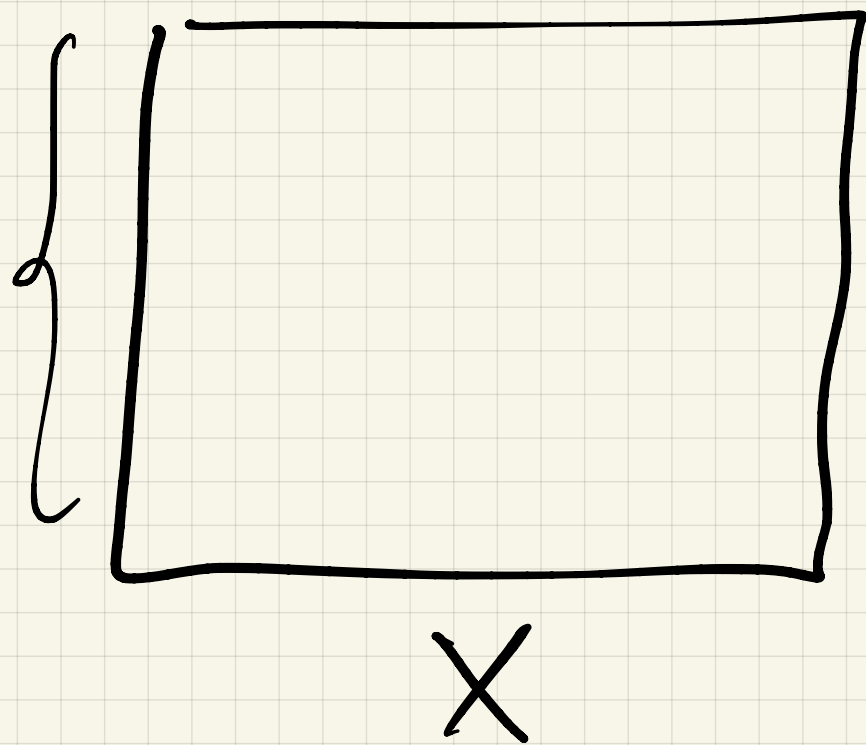
Validation

Why regularisation, four objectives, one score

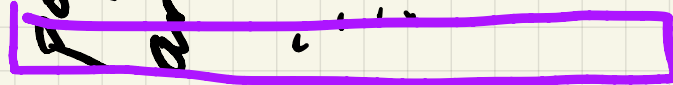
Carlos Cotrini

Weekend 1

1) GDP
y



pop size
area



AI

CS

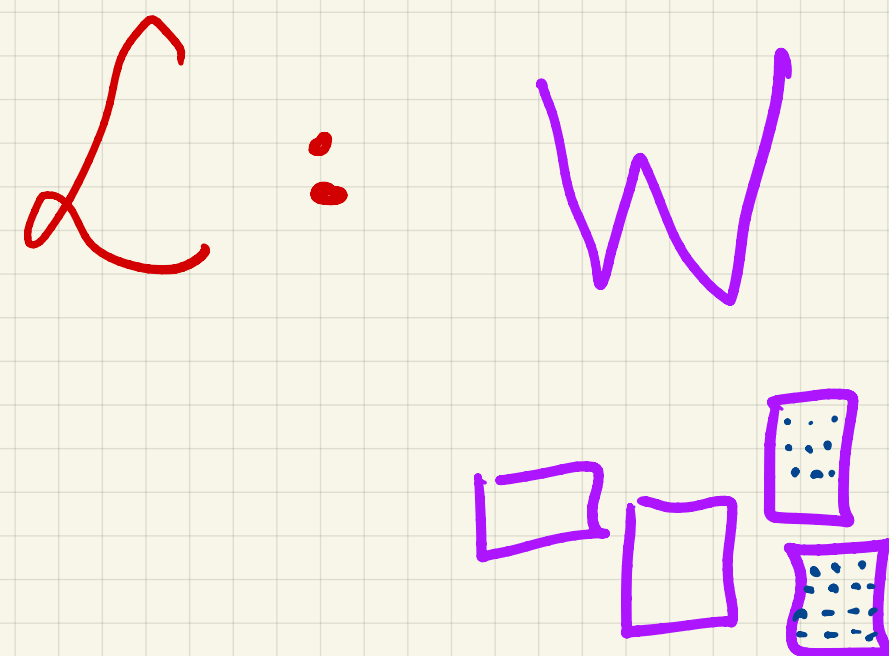
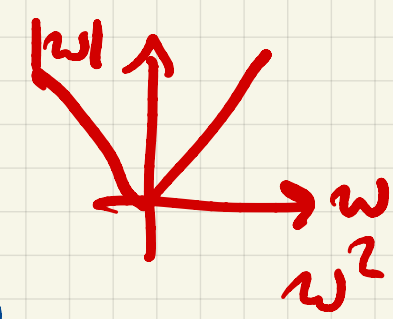
l

$\nabla \bar{e}$

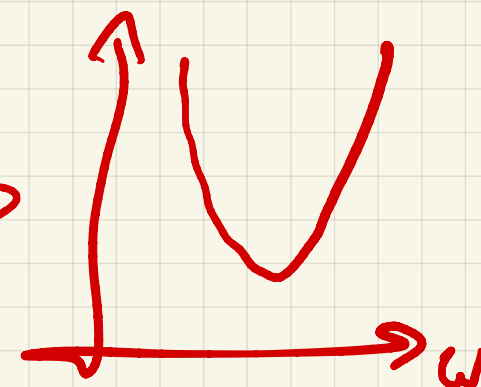
$\bar{e}$

2)

Root: All features have a say.



Reg. term



$$\mathcal{L} + \alpha \times \# \text{ of weights that are } \neq 0$$

can't compute gradients

$$w_1^2 + w_2^2 + w_3^2 + w_4^2 + \dots + w_{\dots}^2$$

$$|w_1| + |w_2| + |w_3| + |w_4| + \dots + 1$$

sol 1
Ridge reg.

sol 2
LASSO

We use linear models!

$$\begin{array}{ccccccc} w_1 & w_2 & & & w_d & & \\ x_1 & x_2 & \dots & x_d & \hat{y} = w_1 x_1 + w_2 x_2 + \dots & & \\ \underbrace{\hspace{15em}} & & & & & & \\ \text{features} & & & & & & \end{array}$$

The diagram illustrates a linear model equation. The weights $w_1, w_2, \dots, w_d$ are written in purple above the features $x_1, x_2, \dots, x_d$. A black brace under the features is labeled "features". A red arrow points from the predicted value $\hat{y}$ to the equation $\hat{y} = w_1 x_1 + w_2 x_2 + \dots$. A purple arrow points from the weight w_d to the same equation.

1

Section 1

Why regularisation

Two numbers

$$R_{\text{emp}}(w) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$$

what you can compute

hope

$$R(w) = \int L(y, \hat{y}) dP(x, y)$$

what you are paid for

Least squares, 203 countries, 203 columns:

0.997 on the data it fitted, -9.41 on the data it had not.

The gap has a price

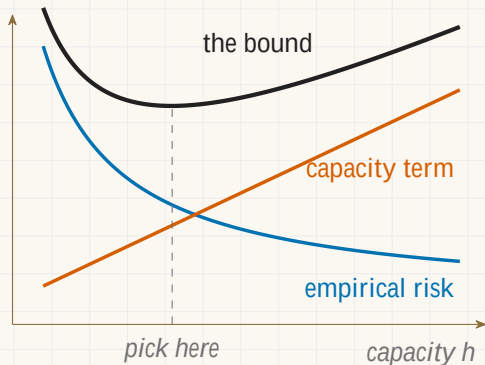
$$\boxed{R(w)} \leq \boxed{R_{\text{emp}}(w)} + \boxed{\Omega\left(\frac{h}{n}\right)}$$

the true risk *what you drove down* *capacity h , sample n*

Zero training error in a large enough family buys nothing.
The second term pays for it.

Structural risk minimisation

- Minimise the sum.
- One α walks the family.
- Large α , small family.



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Section 2

Four objectives

Least squares

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

miss on the data

no penalty

A unique minimiser needs $p < n$.

All 493 columns: 1.000 on train, 0.333 on test.

Ridge

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

miss on the data

+

$$\alpha \sum_{j=1}^p w_j^2$$

the L2 penalty

Smooth at zero, so nothing lands on zero.

Best test **0.592** at $\alpha = 81.1$, keeping **487 of 493**.

Lasso

$$\boxed{\sum_{i=1}^n (y_i - \hat{y}_i)^2} + \boxed{\alpha \sum_{j=1}^p |w_j|}$$

miss on the data *the L1 penalty*

A corner at zero, so coefficients land on it.

Best test **0.599** at $\alpha = 285$, keeping **109 of 493**.

Elastic net

$$\boxed{\sum_{i=1}^n (y_i - \hat{y}_i)^2} + \boxed{\alpha \left(\rho \sum_j |w_j| + \frac{1-\rho}{2} \sum_j w_j^2 \right)}$$

miss on the data *both, split by ρ*

For columns that move together.

Best test **0.592** at $\alpha = 0.811$, keeping **485 of 493**.

The four side by side

Method	Penalty	Zeroes?	Best test	Kept
Least squares	none	no	0.333	493
Ridge	$\alpha \sum w_j^2$	no	0.592	487
Lasso	$\alpha \sum w_j $	yes	0.599	109
Elastic net	both, split by ρ	yes	0.592	485

The three α values are **not on one scale**.

Ridge alone carries no $\frac{1}{2n}$. Compare the scores.

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Section 3

Reading the score

R squared

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- Your miss, over the miss of always predicting $\bar{y}$.
- No floor.

